

Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

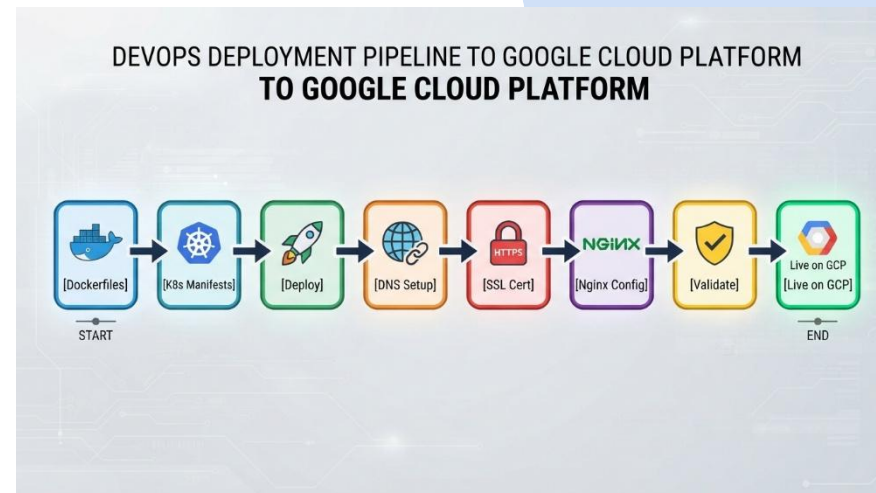
Meeting 9 OllamaModelTester Deployment

Presented by: Mariyan Vasilev Apostolov
Date: 16th of September 2026
Version: 1.0

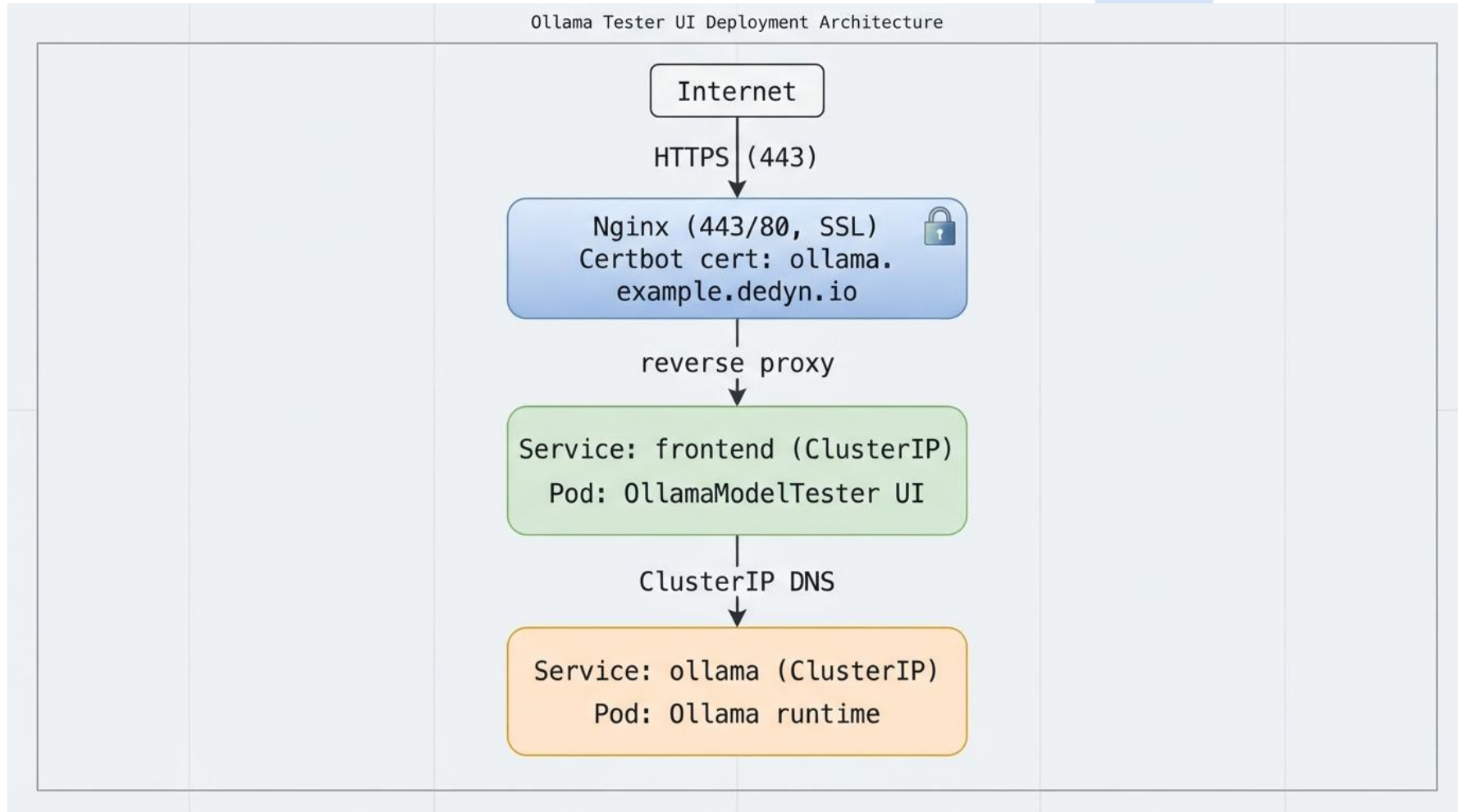
2

Agenda

- Architecture Overview
- Dockerfiles for Each Component
- Kubernetes Manifests – Deployments
- Kubernetes Manifests – Services
- Deploying Components to the Hosted Server
- Domain Creation in Desec.io
- SSL/HTTPS Certificate with Certbot
- Nginx Configuration for SSL/HTTPS
- Inter-Service Communication Validation



Architecture Overview

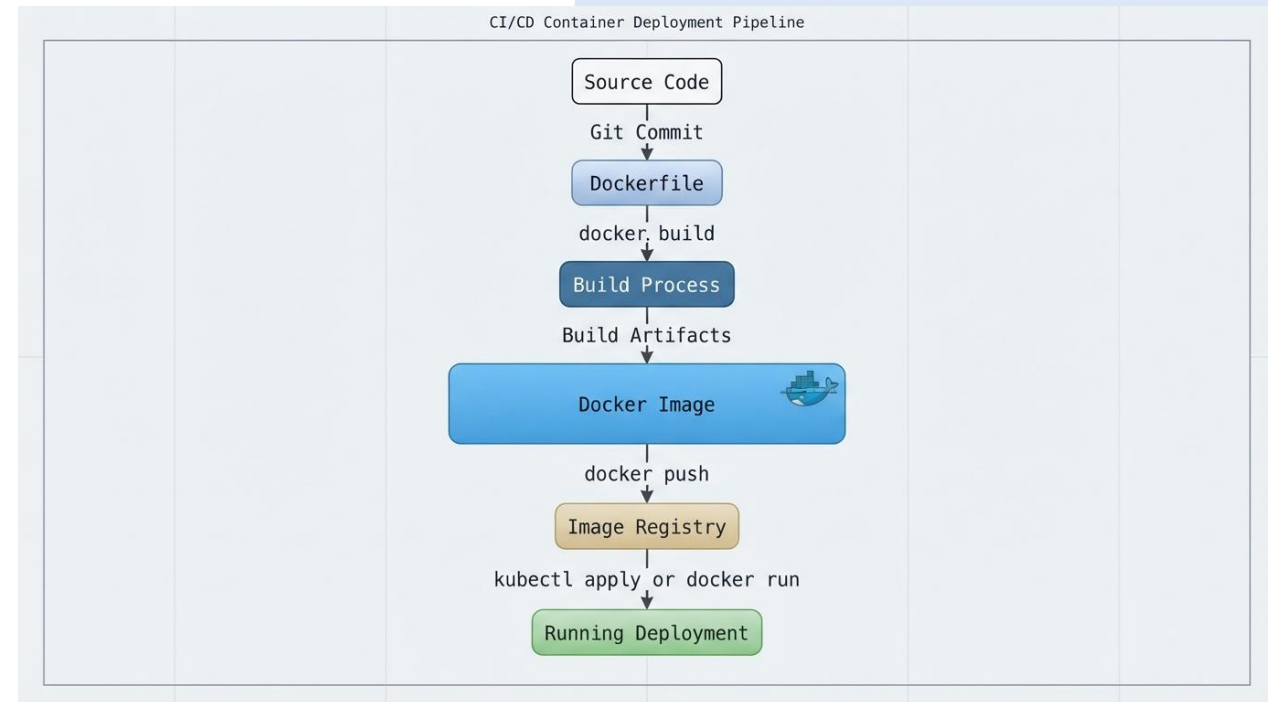


Dockerfiles for Each Component

Package each application component into a portable, reproducible container image.

Components Containerized

Component	Role	Base Image Type
Frontend	User interface	Lightweight web server
Backend API	Business logic	Runtime environment
Ollama Service	Model inference	GPU/CPU runtime
Database	Data persistence	Managed DB image

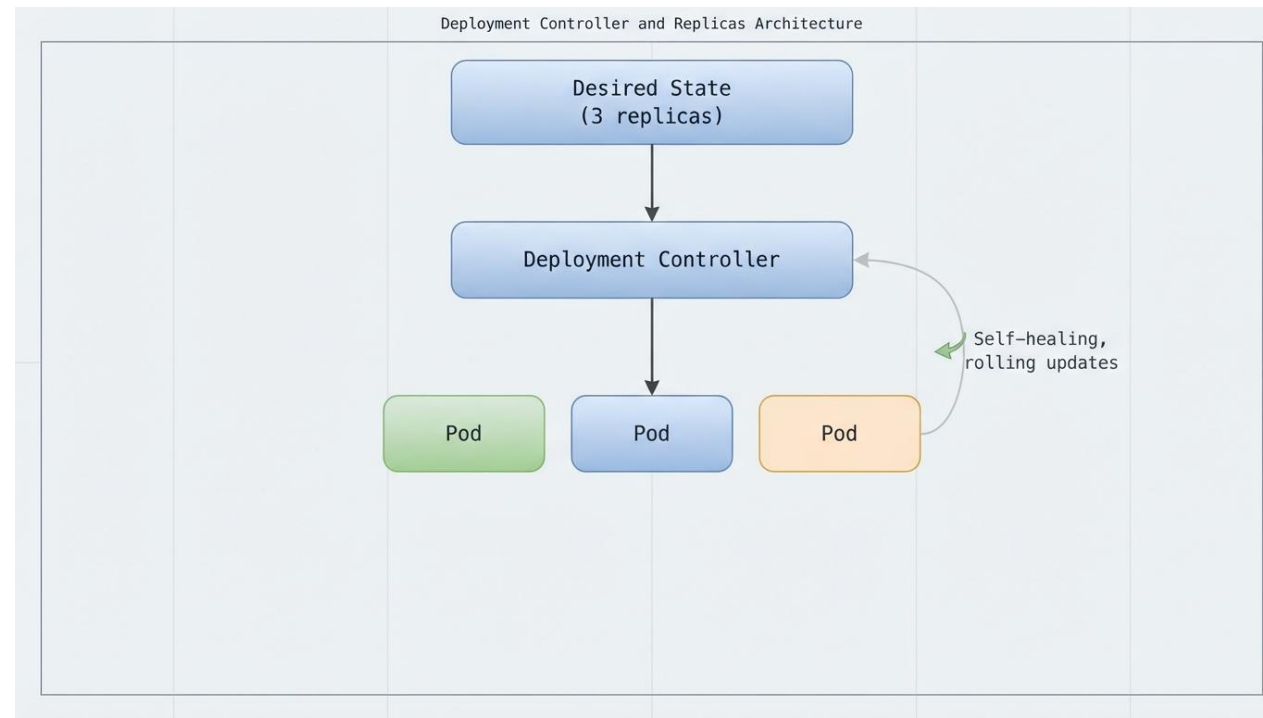


5

Kubernetes Manifests – Deployments

Deployment Manifests (YAML)

- Each component receives a Deployment manifest defining:
- Replicas: Desired pod count
- Container image: Source from registry
- Ports: Exposed container ports
- Environment variables: Configuration injection
- Resource limits: CPU and memory bounds



6

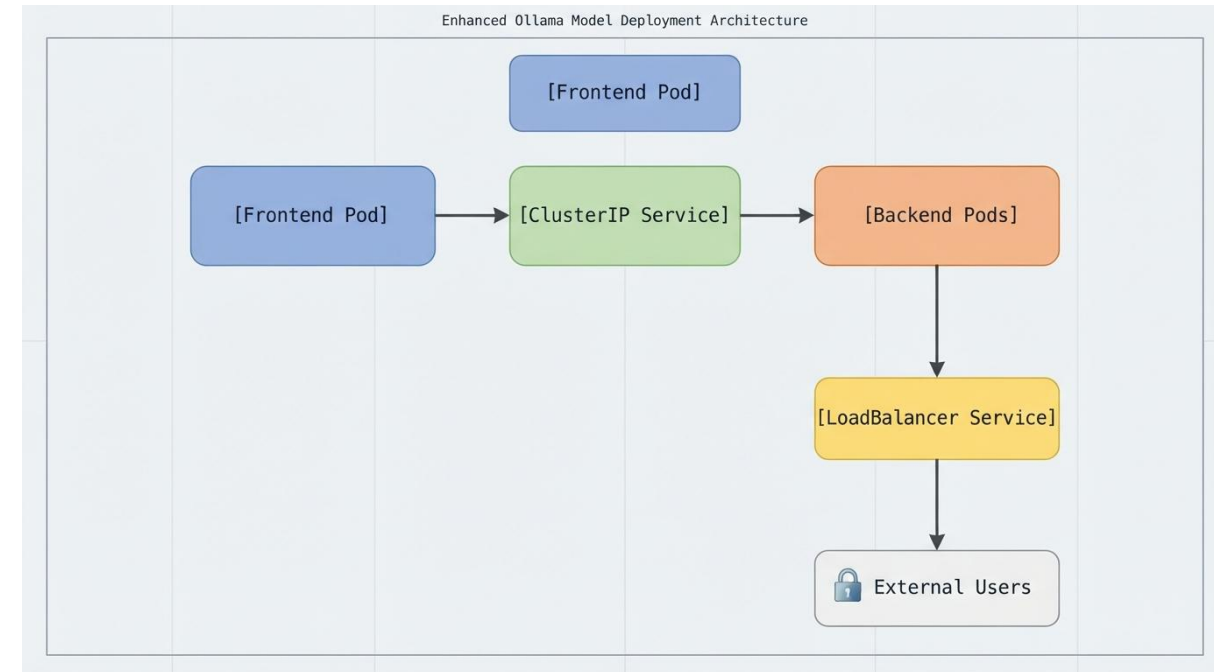
Kubernetes Manifests – Services

Service Manifests (YAML)

Services provide stable networking for ephemeral pods.

Service Types Used:

Type	Purpose	Accessibility
ClusterIP	Internal communication	Within cluster only
NodePort	External access (dev/test)	Node IP + static port

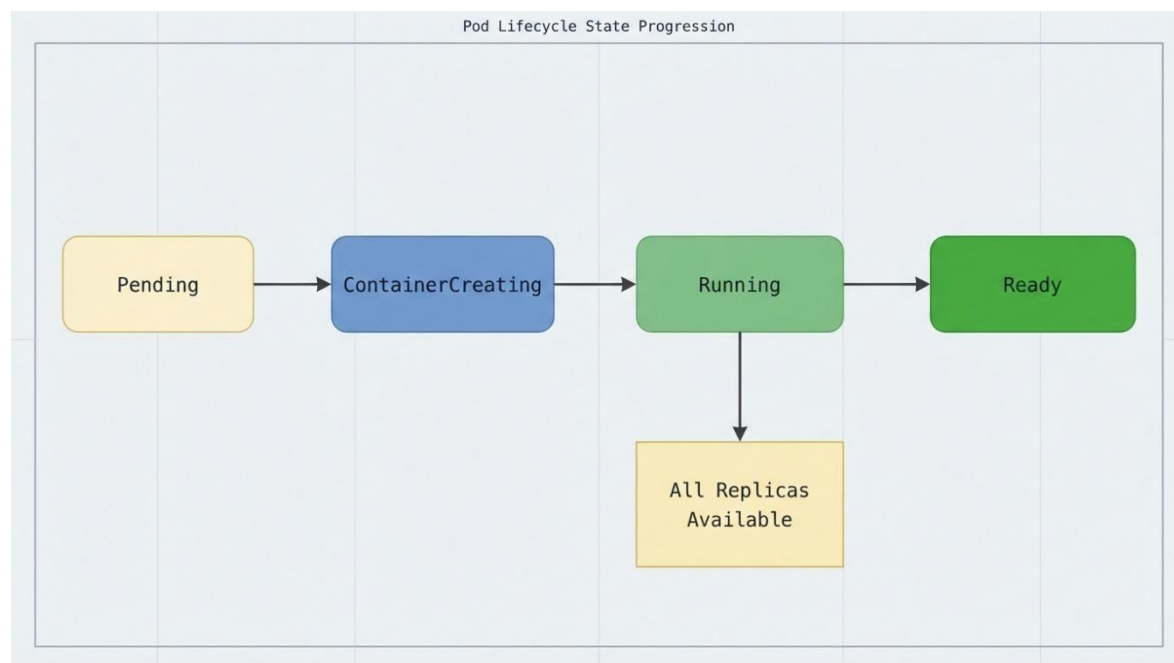


Deploying Components to the Hosted Server

Deployment Execution

Steps performed:

- Apply all manifests to the cluster
- Verify pod status and readiness
- Confirm service endpoints are populated
- Check logs for startup errors



8

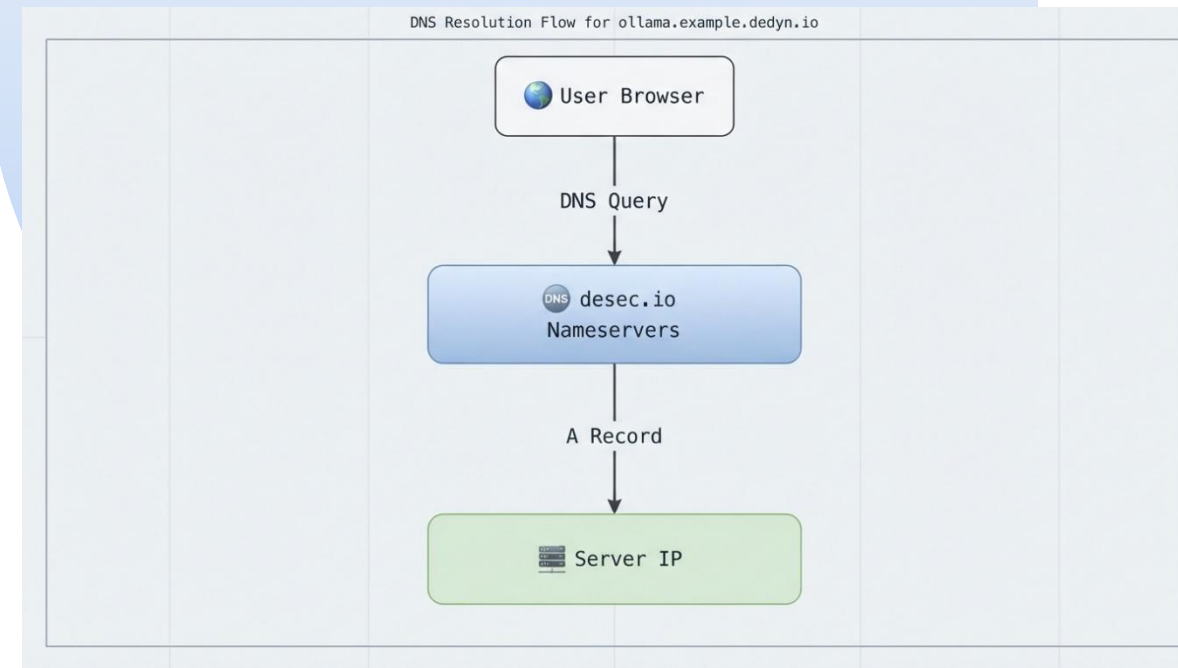
Domain Creation in Desec.io

Domain Management

Desec.io provides free, API-driven DNS hosting.

Steps Taken:

- Create account and log in to Desec.io
- Register the desired domain name
- Configure DNS records:
 - A record: Points domain to server IP
 - CNAME record: Optional subdomain aliasing
- Set TTL values for propagation

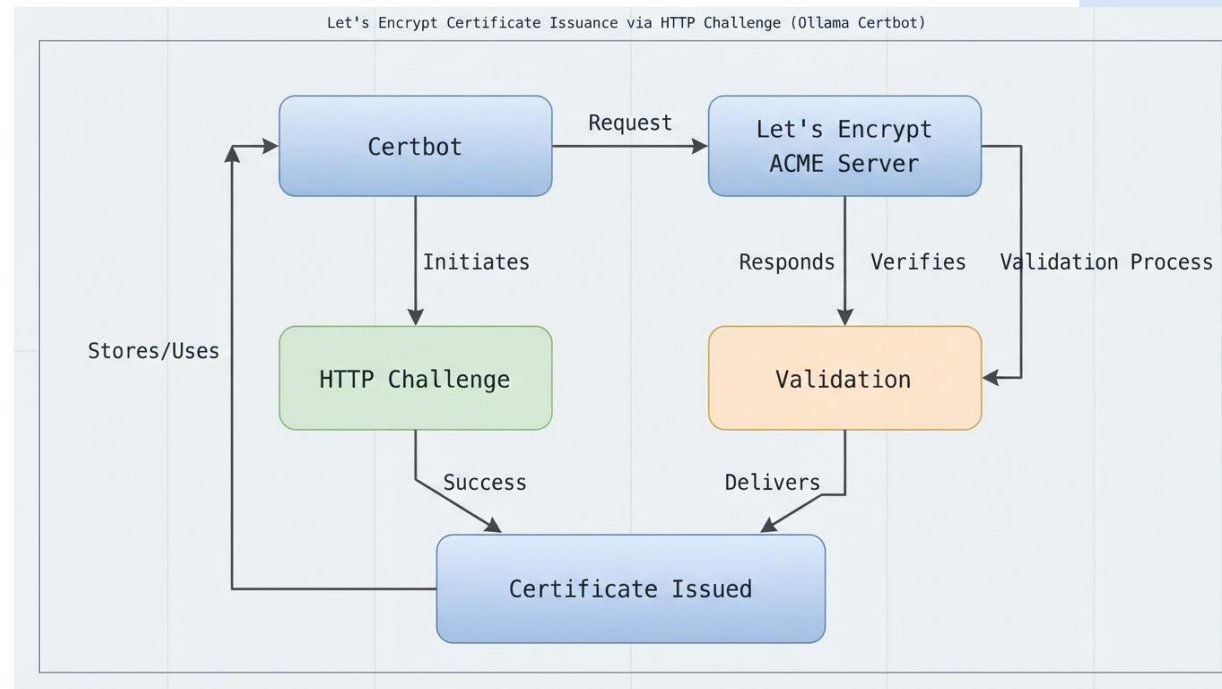


SSL/HTTPS Certificate with Certbot

Purpose: Obtain a free, auto-renewing SSL certificate from Let's Encrypt.

Process:

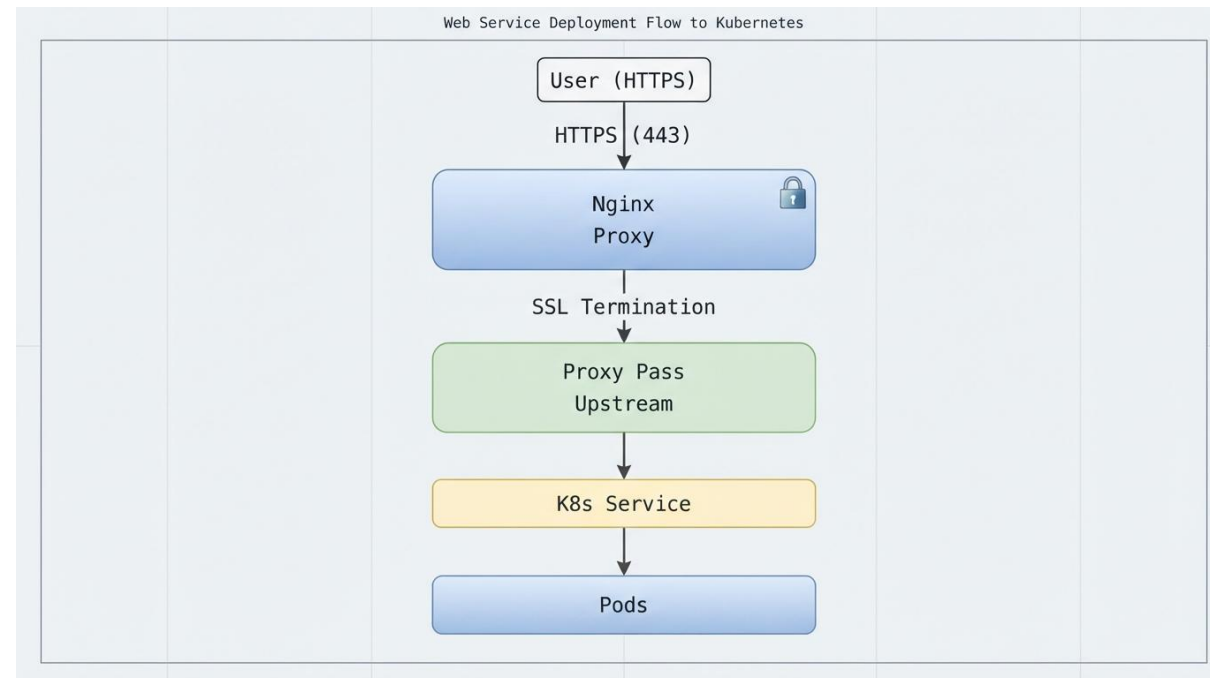
- Install Certbot on the server
- Run Certbot with domain and webroot configuration
- Complete HTTP challenge verification
- Certificate issued and stored locally



Nginx Configuration for SSL/HTTPS

Nginx as Reverse Proxy:

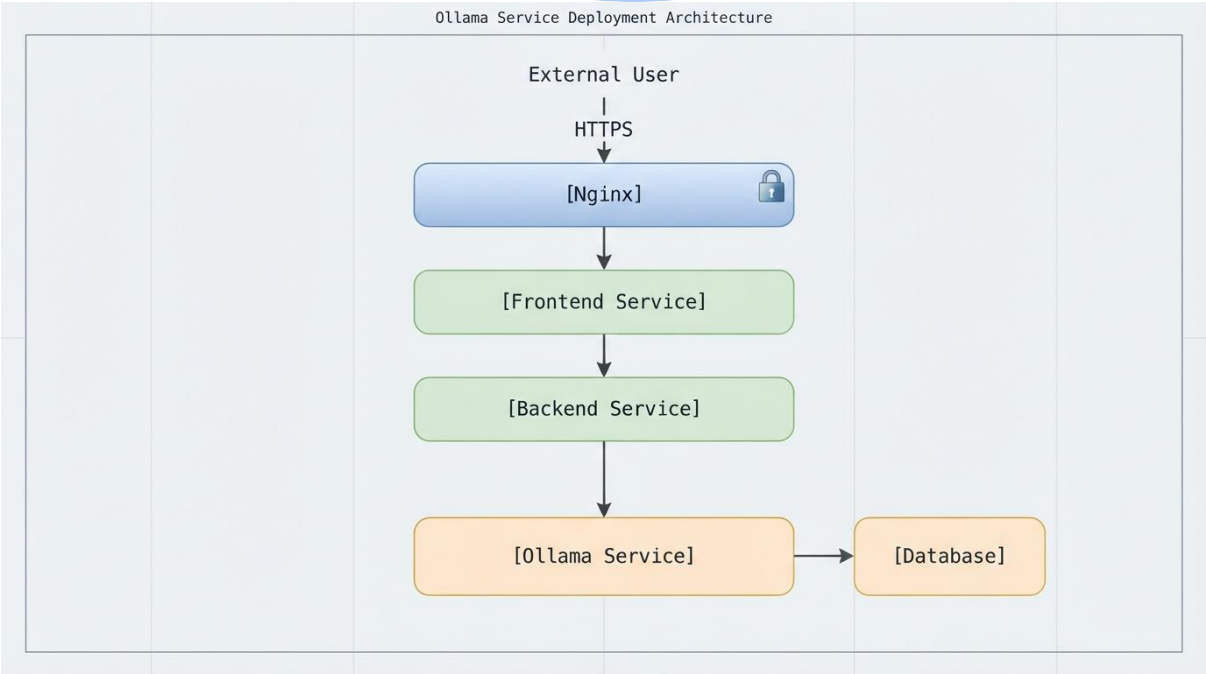
- Install and start Nginx
- Configure server block for the domain
- Point SSL certificate paths to Certbot output
- Set up HTTP → HTTPS redirect
- Configure proxy pass to Kubernetes services



Inter-Service Communication Validation

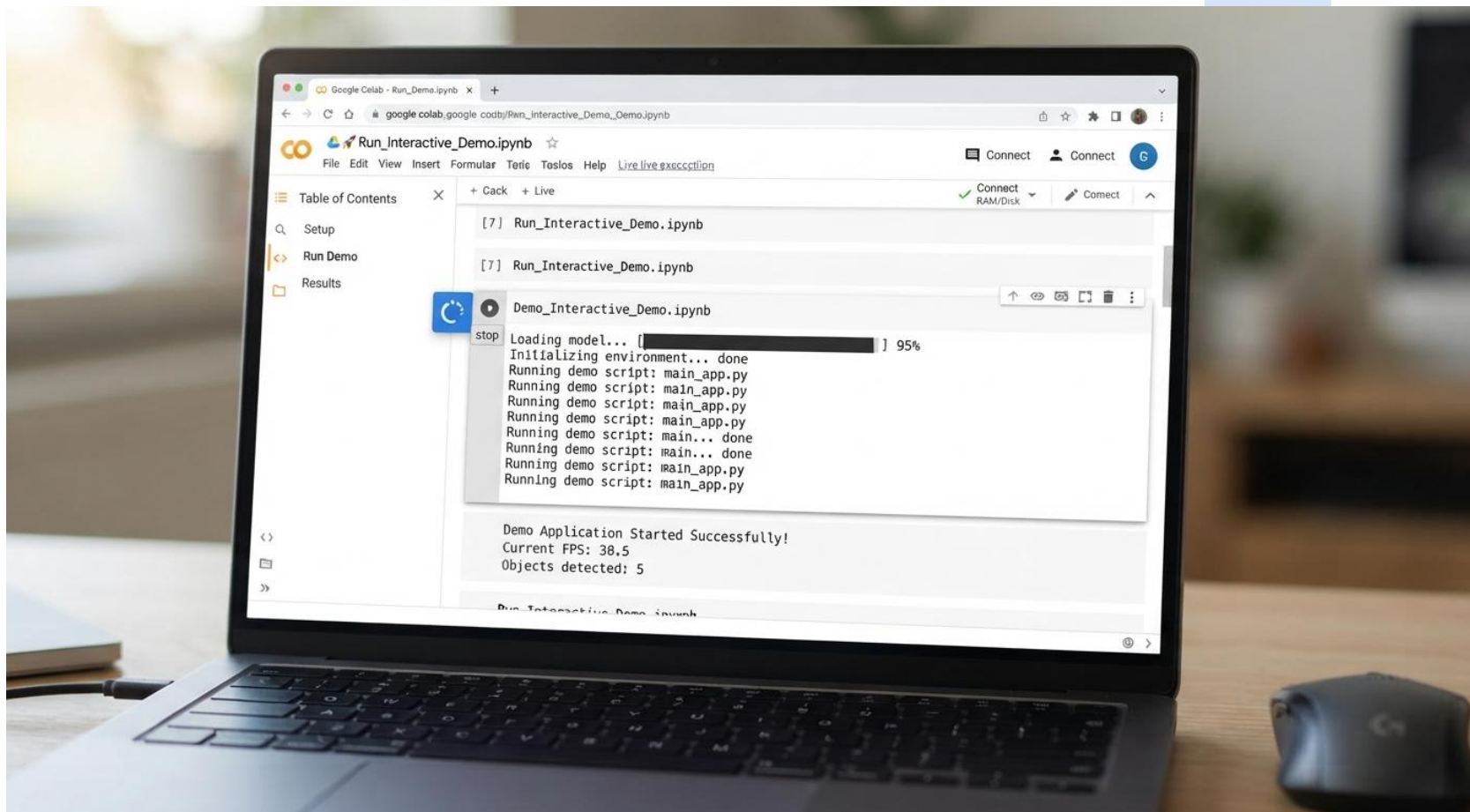
Validation Strategy

Test	Method	Expected Result
Frontend → Backend	Internal HTTP call	200 OK
Backend → Ollama	API request	Model response
Backend → Database	Query execution	Data returned
External → Frontend	HTTPS request	Page loads



12

LIVE DEMO

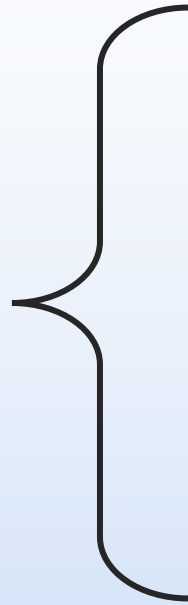


References



- Kubernetes basics – <https://kubernetes.io/docs/tutorials/kubernetes-basics/>
- Start Minikube – <https://minikube.sigs.k8s.io/docs/drivers/docker/>
- Minikube Dashboard – <https://minikube.sigs.k8s.io/docs/handbook/dashboard/>
- Minikube Dashboard commands – <https://minikube.sigs.k8s.io/docs/commands/dashboard/>
- Minikube access apps – <https://minikube.sigs.k8s.io/docs/handbook/accessing/>
- Minikube delete clusters – <https://minikube.sigs.k8s.io/docs/commands/delete/>
- Kubernetes setup clusters on Ubuntu – <https://medium.com/@venkataramarao.n/kubernetes-setup-two-node-cluster-on-ubuntu-20-04-43e2cb38e643>
- Kubernetes service – <https://kubernetes.io/docs/concepts/services-networking/service/>
- DNS Checker – <https://dnschecker.org/>
- Domain management – <https://desec.io/>
- Chmod 600 – <https://chmodcommand.com/chmod-600/>

Thank you



Mariyan Vasilev Apostolov

mariyan.apostolov89@gmail.com